W11 Open Session Notes

1.

X_i: 1 → success ; 0 failure → indep. (n)

$$X = \sum_{i=1}^{n} X_i \sim Bin(n, p)$$

toss a coin '5' times.

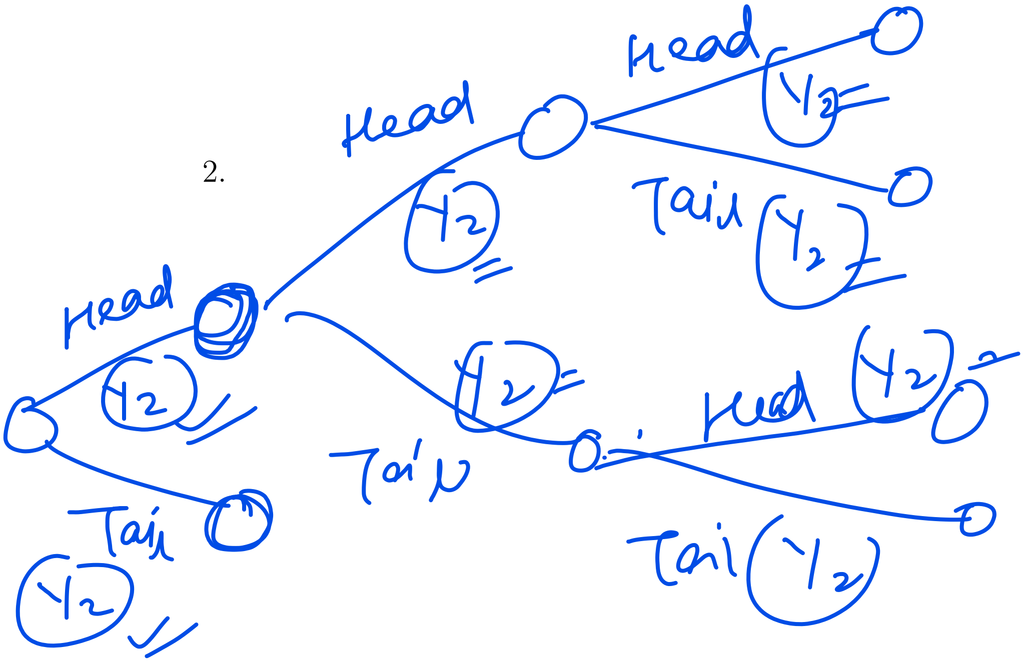
$X \to$ no. of heads ; 0, 1, 2, 3, 4, 5

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

$E(X) = np$; $Var(X) = np(1-p)$

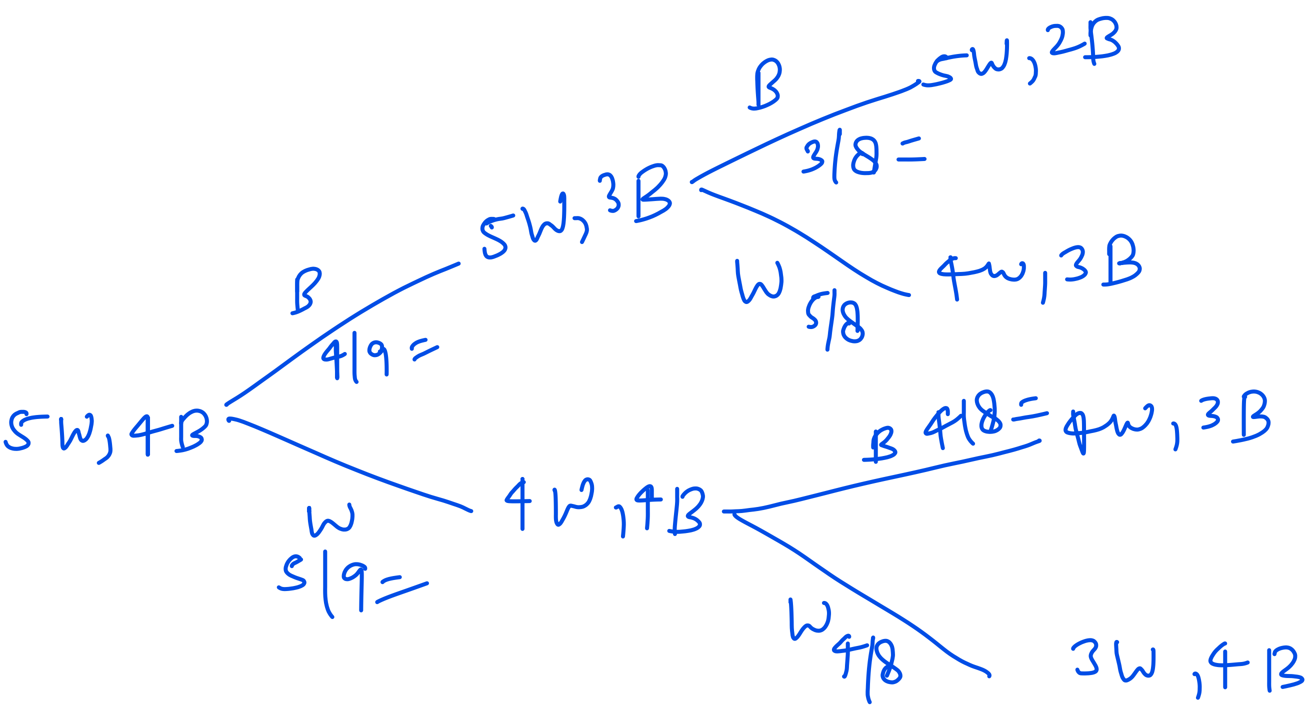
	1st time	2nd time	3rd time	4th	5th
$p \to$	$\frac{1}{2}$	~1/2	1/2	~1/2	~1/2

2.



$p = 1/2$

the prob. of "success" is not same for all trials and "trials" are not indep.



X

↳ ① A fixed no. of trials.

3.

② The success prob. is not same for trial to trial.

Pop^n — 2 groups $\begin{pmatrix} W \\ B \end{pmatrix}$

→ We sample from groups of items when we're interested in only one group.

X: no. of success out of total no. of items chosen.

N → items (in total), randomly select 'n' items without replacement

'm' → 1st type

'$N-m$' → 2nd type

X → no. of items of type I

$$P(X=x) = \frac{\binom{m}{x}\binom{N-m}{n-x}}{\binom{N}{n}}; \quad x = 0, 1, \ldots, n.$$

$N = 7$ balls

$4W$ ($N-m = 7-m$), $3B$ (m), $n = 2$

To draw two balls without replacement.

$X \rightarrow$ no. of black ball selected.

$n = 3$

$i = 0, 1, 2, 3$

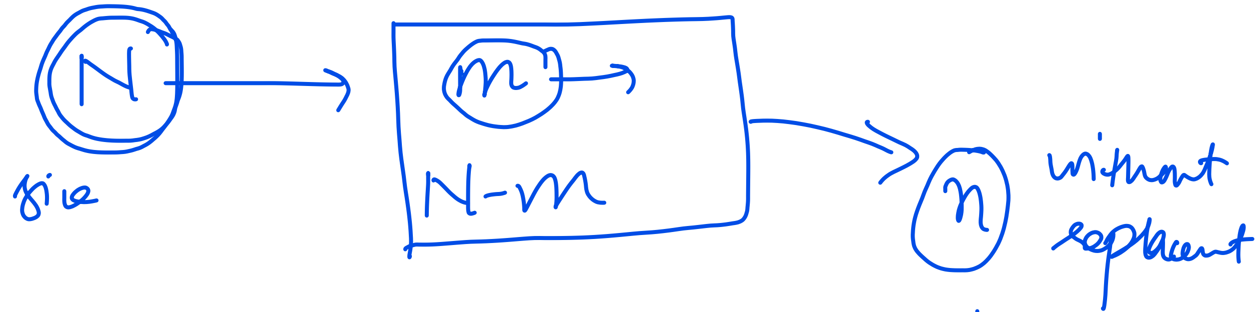
$$P(X = 1) = \frac{\binom{3}{1 = i}\binom{4}{1 = n-i}}{\binom{7}{2}} = \frac{\text{fav cases}}{\text{Total cases}}$$

$P(X = i) =$

$$P(X = 0) = \frac{\binom{3}{0}\binom{4}{2}}{\binom{7}{2}}$$

$$P(X = 2) = \frac{\binom{3}{2}\binom{4}{0}}{\binom{7}{2}} = \frac{\binom{3 \,(m)}{i}\binom{4 \,(N-m)}{2-i}}{\binom{7 \,(N)}{2 \,(n)}}$$

5.



$X \to$ no. of success out of total 'n'

$$\underbrace{P(X=i)} = \frac{\binom{m}{i}\binom{N-m}{n-i}}{\binom{N}{n}} \; ; \; \underbrace{i=0,1,\ldots,n}.$$

Eg:- '52' cards ; $n=5$

$X \to$ no. of aces in the random sample of 5 cards.

$$\underbrace{P(X=i)} = \frac{\binom{4}{i}\binom{48}{5-i}}{\binom{52}{5}} \; ; \; i=0,1,2,\ldots,4.$$

Expectation of hypergeometric dis^n

$$E(X) = \frac{n \cdot m}{N} = n \cdot \frac{m}{N}$$ prop.

$X \sim Bin(n, p)$ → prob of success ; proportion.

$E(X) = n \cdot p$

* variance of hypergeometric dis^n

$$Var(X) = n \cdot \frac{m}{N} \cdot \frac{N-m}{N} \cdot \left(\frac{N-n}{N-1}\right)$$

$$\left(1 - \frac{m}{N}\right)$$

$n = 1$

$$Var(X) = n \cdot p \cdot (1-p)$$

$$\frac{N-n}{N-1} = \frac{\cancel{N}\left(1 - \frac{n}{N}\right)}{\cancel{N}\left(1 - \frac{1}{N}\right)}$$ $\frac{1}{N} \to 0$

if $N >> n$

(I)

7.

$$Var(X) = n \cdot \frac{m}{N} \cdot \left(1 - \frac{m}{N}\right) \cdot \left(\frac{N-n}{N-1}\right)$$

$n=1$ → drawing a single sample?

↳ Bernoulli

$$= \frac{m}{N}\left(1-\frac{m}{N}\right) = p(1-p) \longrightarrow Var(Ber)$$

$$E(X) = n \cdot \frac{m}{N} = \frac{m}{N} = p \to Ber$$

$n = N$ → whole population is sampled.

$\underline{Var(X) = 0}$

$N \gg n \rightarrow$ Hypergeometric → Binomial

~~usecols · 1:2 → 'admin~content'~~

8.

20 fruit $= N$

16 → $N-m$

4 → m

→ $n = 5$ fruits

$Y \to$ no. of rotten fruits

$$P(Y=2) = \frac{\binom{4}{2}\binom{16}{3}}{\binom{20}{5}} \; ; \; y = 0, 1, 2, 3, 4$$

$N = 52$

39

13 spades

→ $n = 8$ cards

$$P(X<3) = P(X=0) + P(X=1) + P(X=2)$$

8

$$P(X=i) = \frac{\binom{13}{i}\binom{39}{8-i}}{\binom{52}{8}}$$

Poisson distn:

→ gives the probability of a "no. of events occuring in a fixed interval of time or space".

Eg:- Arrivals of customers at a service center during a specific time period

→ the no. of customers arriving at a bank per hour

Eg2:- Phone calls at a call center.

:- the no. of phone calls received by a call center in a given hour.

Eg 3: <u>Accidents on a Road:</u>

↳ the no. of accidents occuring at a particular intersection of the road at a given period

Eg 4: → No. of spelling mistakes on a page of a book.

X :– no. of times an event occurs in an interval of time (or space).

Note :– Bin (n, p)

n ↓ no. of trials >> large.

p ↳ p is very very small

$np \longrightarrow \lambda$ (constant)

<u>Poisson distn.</u>

$np = \lambda \Rightarrow p = \dfrac{\lambda}{n}$

Prob. mass functions of poisson dist^n

The dist^n, with an average no. of λ events per interval, is defined as poisson discrete random variable, $\underbrace{X \sim \text{Poisson}(\lambda)}$, with the p.m.f given by

$$P(X=x) = \frac{e^{-\lambda}\,\lambda^{x}}{x!} \; ; \; x = 0, 1, 2, \ldots$$

$\lambda \to$ parameter

$e \to$ mathematical constant

$X \to$ no. of events per time interval

$$\boxed{E(X) = \lambda, \quad Var(X) = \lambda}$$

##

Examples: (Poisson distn)

12. (1) No. of vehicles passing through a traffic intersection in a fixed interval of time.

(2) no. of people withdrawing money from a bank in a fixed interval of time

Space:

(1) No. of typos in a book.

(2) no. of defects per meter in a roll of cloth

(3) no. of defects in a wire cable of finite length.

Eg:- The no. of dogs that are killed on
13. particular intersection of road in a city on any day follow Poisson(0.4)

(1) Prob. that exactly two dogs are killed on a given day

$X \sim \text{poisson}(\lambda)$

$$P(X=x) = \frac{e^{-\lambda}\lambda^{x}}{x!}; \quad x=0,1,2,\ldots$$

$$P(X=2) = \frac{e^{-0.4}(0.4)^2}{2!}$$

$$P(X=3) = \frac{e^{-0.4}(0.4)^3}{3!}$$

P(less than 4 dogs are killed) = $P(X<4)$

$X \sim P(0.4) \longrightarrow \lambda$

$$\frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$= P(X=0) + \underbrace{P(X=1)} + P(X=2) + P(X=3)$$

14.

$$= \frac{e^{-0.4}(0.4)^0}{0!} + \frac{e^{-0.4}(0.4)^1}{1!} + \frac{e^{-0.4}(0.4)^2}{2!}$$

$$+ \frac{e^{-0.4}(0.4)^3}{3!}$$

$$\frac{e^{-\lambda}\lambda^x}{x!}$$

Q: The no. of dogs that are killed on a particular intersection of a road in any one day follows $\underbrace{\text{poisson}(0.4)}$. $\longrightarrow \lambda = 0.4$

① Prob. that exactly two dogs are killed on a given day:

$$P(X=2) = \frac{e^{-0.4}(0.4)^2}{2!}$$

② Prob. that exactly two dogs are killed over a $\underbrace{\text{2-day}}$ period:

$$\lambda' = \underbrace{2 \times 0.4} = 0.8.$$

$$P(x=2) = \frac{e^{-\lambda}\lambda^x}{x!} = \frac{e^{-0.8}(0.8)^2}{2!}$$

15.

(3) Prob. that exactly two dogs are killed over a period of 5 days.

new $\lambda = 5 \times 0.4$
$= 2$

$$P(X=2) = \frac{e^{-2}\,2^2}{2!}$$

in an hour $= (\lambda = 2)$

next four hour $= (\lambda = 8)$

$$P(X=3) = \frac{e^{-8}\,8^3}{3!}$$

The no. of persons arriving per hour in an

16. Insurance company $\sim$ Poisson (6) $\to \lambda$

Prob. that exactly 7 people will arrive in an hour

$$P(X=7) = \frac{e^{-6}\, 6^{7}}{7!}$$

$X \sim P(\lambda)$: $E(X^2) = 6$

$E(X) \to ?$

$$\begin{cases} E(X) = \lambda \\ Var(X) = \lambda \end{cases}$$

$$\Rightarrow E(X^2) - (E(X))^2 = \lambda$$

$$\Rightarrow 6 - \lambda^2 = \lambda$$

$$\Rightarrow \lambda^2 + \lambda - 6 = 0$$

$$\Rightarrow \lambda^2 + 3\lambda - 2\lambda - 6 = 0$$

17. $\Rightarrow \lambda(\lambda+3) - 2(\lambda+3) = 0$

$$\Rightarrow (\lambda+3)(\lambda-2) = 0$$

$$\Rightarrow \lambda = \cancel{-3} \text{ or } 2$$

$$\frac{e^{-\lambda}\lambda^{x}}{x!} = P(X = x)$$

$\lambda \Rightarrow E(X) = \lambda$

$\Rightarrow V(X) = \lambda$

$= \sqrt{\lambda}$

$$P(X=3) = \frac{2}{3} P(X=4)$$

$$\Rightarrow \frac{\cancel{e^{-\lambda}} \lambda^{\cancel{3}}}{3!} = \frac{2}{3} \cdot \frac{\cancel{e^{-\lambda}} \lambda^{\cancel{4}}}{4!}$$

$$\Rightarrow \frac{3}{3!} = \frac{2 \cdot \lambda}{4!}$$

$$\Rightarrow \frac{3}{\cancel{3!}} = \frac{\cancel{2}\lambda}{\underset{2}{\cancel{4}} \times \cancel{3!}}$$

$$\Rightarrow 3 = \frac{\lambda}{2}$$

$$\Rightarrow \lambda = 6$$

$$E(X) = \lambda$$

$$V(X) = \lambda = 6$$

$$SD(X) = \sqrt{6}$$

$$F(a) = P(X \le a)$$

CDF → cumulative distr fn

$$\left.\begin{array}{llllll} X: & x_1 & x_2 & \cdots & x_5 & , \\ P(X=a): & p_1 & p_2 & \cdots & p_5 & \end{array}\right\} \underline{\text{PMF}}$$

$P(X = x_1) = p_1$; $P(X = x_3) = p_3$

$P(\underbrace{X \le x_3}) \to$ accumulates the prob. till x_3

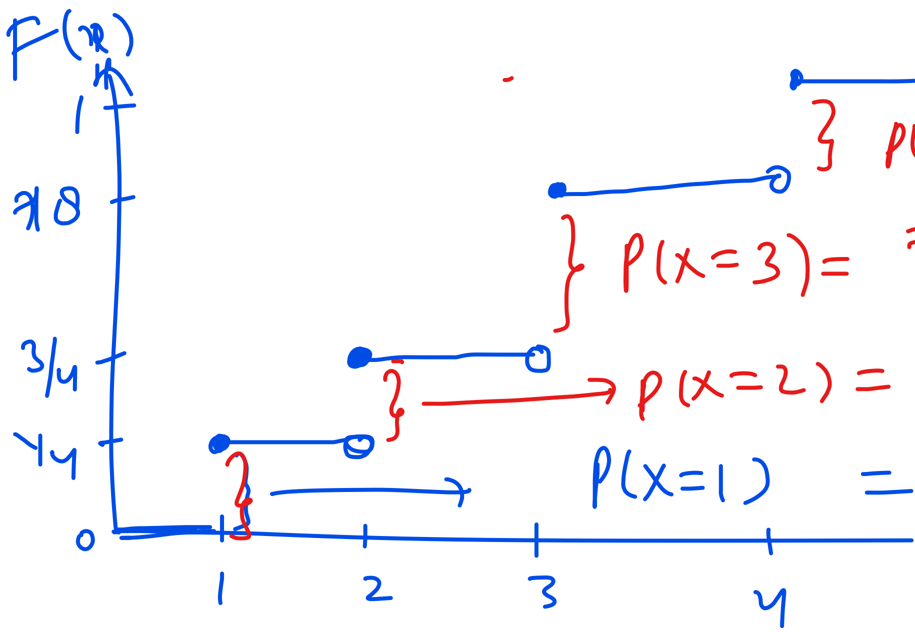
$$= P(X=x_1) + P(X=x_2) + P(X=x_3)$$

X:	1	2	3	4
$P(X=a)$:	$1/4$	$1/2$	$1/8$	$1/8$

X:	1	2	3	4
$P(X)$:	$1/4$	$1/2$	$1/8$	$1/8$

20.

$$F(a) = \begin{cases} 0 & ; a < 1 \\ 1/4 & ; 1 \le a < 2 \\ 1/4 + 1/2 = 3/4 & ; 2 \le a < 3 \\ 7/8 & ; 3 \le a < 4 \\ 1 & ; 4 \le a \end{cases}$$

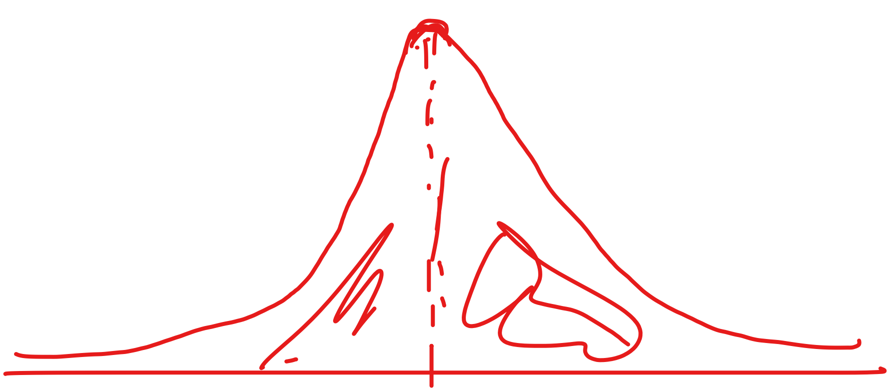


$P(X=4) = 1 - 7/8 = 1/8$

$P(X=3) = 7/8 - 3/4$

$P(X=2) = 3/4 - 1/4 = 1/2$

$P(X=1) = P(X=1) - P(X \le 1)$



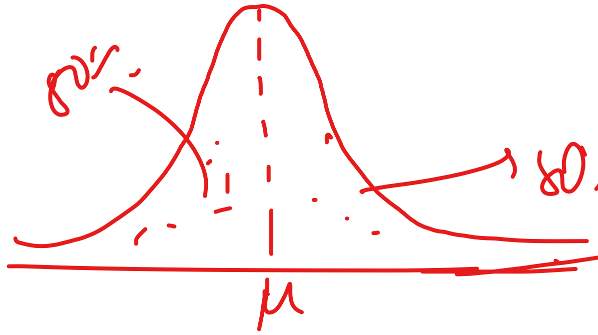
n > large.

n ≥ 30 → large sample

normal distn

7:30

21.



n → large samples → Normal distn

prob. of owing a Creta = 4.5% = 0.045

Prob. that 12th person being interviewed is the 2nd one who own a Creta.

$\underbrace{(1 \; . \; . \; . \; . \; - \; 11 \text{ people})}_{\text{1st one}} \quad (12\text{th person}) \rightarrow 2\text{nd}$

$$= \binom{11}{1}(0.045)^{1}(1-0.045)^{10} \times (0.045)$$

$P(x=0) = 0.15$

$$\binom{3}{0} p^{0}(1-p)^{3} = 0.15$$

$$(1-p)^{3} = 0.15$$

$$1-p = (0.15)^{1/3}$$

22.

$$\binom{3}{3} p^3 (1-p)^0 = 0.25$$

$$p^3 = 0.25$$

$$\Rightarrow p = (0.25)^{1/3}$$

$$V(X) = E(X^2) - (E(X))^2$$

$$E(X) = \sum x\, p(x)$$

$$E(X^2) = \sum x^2\, p(x)$$

$$np = 1.6$$

$$3p = 1.6$$

$$p = \frac{1.6}{3} = 0.533$$

$$3 \times (0.533) \times (1 - 0.533$$

$$\cancel{3 \times (0.8}$$

23.

24.

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35.